Teacher Wu Kuan | See It at a Glance #29 — Sums of Odd Numbers Make Squares

English lecture notes · YouTube Short MJhcFA03xU8

1. The trap question

1 + 3 + 5 + … + 19 = ?

2. The method

• Each new odd number wraps one L-shaped layer around a square.
• Sum of the first n odd numbers = n^2.
• 1 to 19 has 10 odd numbers → $10^2 = 100$.
• General arithmetic series: (first + last) × number of terms ÷ 2.

3. Parent check method

Have your child build it with coins or blocks: 1, then 3 around it, then 5. Ask what shape appears each time.

Practice

1. Find 1 + 3 + 5 + … + 29.
2. Start at 1 and keep adding consecutive odd numbers until the total is 144. How many odd numbers were added, and what was the last one?
3. Find 2 + 4 + 6 + … + 40.
4. Find 11 + 13 + 15 + … + 39.
5. Build squares layer by layer: 1 block, then 3, then 5, … You have 300 blocks. What is the largest complete square you can make, and how many blocks are left?

Answers

1. 225 (15 odd numbers → 15^2)
2. 12 numbers; the last is 23
3. 420 (first 20 odds = 400, plus 1 for each of 20 terms)
4. 375 (15 terms; (11 + 39) × 15 ÷ 2)
5. 17 × 17 (289 blocks); 11 left over

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